QATIPv3 Azure Lab2
Creating a virtual machine with Terraform

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Lab Objectives

In this lab you will:

- Review the Terraform registry documentation relating to Azure Windows Virtual Machines
- Deploy and then modify an Azure VM
- Destroy the deployment

Teaching Points

This lab takes you through using Terraform to create a virtual machine in Azure. A virtual machine cannot exist in isolation however; it must have a virtual nic which must be on a subnet. The subnet must be part of a virtual network. All these resources must be part of a resource group.

Before you begin

1. Ensure you have completed Lab0 before attempting this lab.
2. In the IDE terminal pane, enter the following command…

   cd c:\azurelabs\lab2

3. This shifts your current working directory to lab2. Ensure all commands are executed in this directory
4. Close any open files and use the Explorer pane to navigate to and open the lab2 folder.
5. Open the provided empty main.tf file

Solution

The solution to this lab can be found in C:\azurelabs\solutions\lab2. Try to use this only as a last resort if you are struggling to complete the step-by-step processes.

Task1 Review the Terraform Registry

1. Review Terraform documentation (Version 4.16.0 is used throughout this course)

   Azure Provider
   Azure Windows Virtual Machine
   Azure Network Interface
   Azure Subnet

Task2 Deploy the resources

1. Navigate to Azure Windows Virtual Machine

2. Scroll down to the ‘Example Usage’ section and copy the sample code into main.tf

3. Insert a new line in the provider block and add your subscription details;

lab2 > main.tf > resource "azurerm_virtual_network" "example" >

```
provider "azurerm" {
  features {}
  subscription_id = "your subscription id here"
}
```

4. Change the resource group name to “rg2-{suffix}”;

```
resource "azurerm_resource_group" "example" {
  name     = "RG2-120526-1"
  location = "West Europe"
}
```

5. Change the vm name to “VM1”

```
resource "azurerm_windows_virtual_machine" "example" {
  name                = "VM1"
  resource_group_name = azurerm_resource_group.example.name
```

6. Change the vm size to “Standard_B2s”;

```
size                = "Standard_B2s"
```

7. Run **terraform init** ensuring you are in the correct working directory (C:\azurelabs\lab2)
8. Run **terraform plan** and review the proposed changes
9. Run **terraform apply** typing **yes** when prompted.

10. Switch to the Azure Portal to confirm the creation of a resource group, vnet, sub-net and virtual machine
11. Return to the terminal and enter **terraform show.** Scroll up and note the last few digits of the id of the newly created virtual machine (yours will differ)…

```
user_data                                   = null
virtual_machine_id                          = "f166ce8e-a238-43a0-b502-4db81919537e"
virtual_machine_scale_set_id                = null
```

Task3 Modify your deployment

1. Return to the IDE, edit **main.tf** by adding the following tags section;

```
tags = {
  Lab-Edit = "firstChange"
}
```

```
    version    = "latest"
  }

  tags = {
    Lab-edit = "firstChange"
  }
}
```

2. Run **terraform plan** and note the proposed changes ..

   Plan: 0 to add, 1 to change, 0 to destroy.

   This indicates that the existing VM will be updated in-situ.

3. Run **terraform apply**, review the proposed changes and type **yes** when prompted
4. Return to the Azure Portal and refresh the view
5. Review the virtual machine and verify that the tag has been added

   Tags (edit) : Lab-Edit : firstChange

6. Return to the terminal and enter **terraform show.** Scroll up and note the last few digits of the id of the virtual machine. This is the same value as in Task2, indicating that the change was applied to the existing virtual machine.

```
user_data                                   = null
virtual_machine_id                          = "f166ce8e-a238-43a0-b502-4db81919537e"
virtual_machine_scale_set_id                = null
```

7. Edit the main.tf file to change the sku value, changing the vm operating system from 2016-Datacenter to 2022-Datacenter;

```
  source_image_reference {
    publisher = "MicrosoftWindowsServer"
    offer     = "WindowsServer"
    sku       = "2022-Datacenter"
    version   = "latest"
```

8. Run **terraform plan** and note the proposed changes ..

   Plan: 1 to add, 0 to change, 1 to destroy.

   This indicates that the existing VM will be replaced.

9. Run **terraform apply**, review the proposed changes and then type **yes** when prompted.
10. Return to the Azure Portal.
11. Refresh the Virtual Machine dashboard to confirm the updates.
12. Return to the terminal and enter **terraform show.** Scroll up and note the last few digits of the id of the virtual machine. This is a different value from Task2, indicating that the change of Operating system required the deletion of the original virtual machine and the creation of a replacement.

```
user_data                          = null
virtual_machine_id                 = "09d45c71-1d0c-4463-8983-4326993cedc3"
virtual_machine_scale_set_id       = null
```

13. **Takeaway**: You need to be aware that some resource attributes are immutable, and some can be updated without the resource needing to be destroyed/replaced. In this case, modifying the machine tags can be done without recreating the virtual machine, whereas changing the Operating system would require destruction and recreation.

Task4 Lab Clean-up

1. Run `terraform destroy` review the output and type `yes`

***** Congratulations, you have completed this lab *****